

الورقة الامتحانية
لامتحان الشهادة الثانوية الأزهرية - نموذج تجريبي
للعام الدراسي ١٤٤٧ هـ - ٢٠٢٥ / ٢٠٢٦ م

المادة/ الفيزياء (لغات) - (تجريبي)	
القسم العلمي	زمن الإجابة: ثلاث ساعات

السؤال	الدرجة بالأرقام	اسم المصحح ثلاثيًا
الأول		
الثاني		
الثالث		
الرابع		
الخامس		
المجموع		

عدد الصفحات (١٦) صفحة
و على الطالب مسئولية المراجعة
والتأكد من ذلك قبل تسليم الكراسة

الرقم السري

مجموع الدرجات (مكتوبًا بالحروف):

اسم المراجع العددي ثلاثيًا: اسم المراجع الفني ثلاثيًا:

الأزهر الشريف - قطاع المعاهد الأزهرية - الإدارة المركزية للامتحانات وشؤون الطلاب و الخريجين

عدد الصفحات (١٦) صفحة
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المادة/ الفيزياء (لغات) - (تجريبي)

الرقم السري

نموذج ثانوية الأزهرية تجريبي

اسم الطالب كاملاً:

رقم الجلوس: اسم المعهد:

التاريخ: / /

تعليمات هامة

عزيزي الطالب / عزيزتي الطالبة:

- اقرأ السؤال بعناية، وفكر فيه جيداً قبل البدء في الإجابة عليه.
 - أجب عن جميع الأسئلة ولا تترك أي سؤال دون إجابة.
 - عند إجابتك عن الأسئلة المقالية، أجب فيما لا يزيد عن المساحة المحددة لكل سؤال.
 - عند إجابتك عن أسئلة الاختيار من متعدد (إن وجدت)، ظلل الدائرة ذات الرمز الدال على الإجابة الصحيحة فقط تظليلاً كاملاً.
- مثال: الإجابة الصحيحة (d) مثلاً

(a)

(b)

(c)

(d)

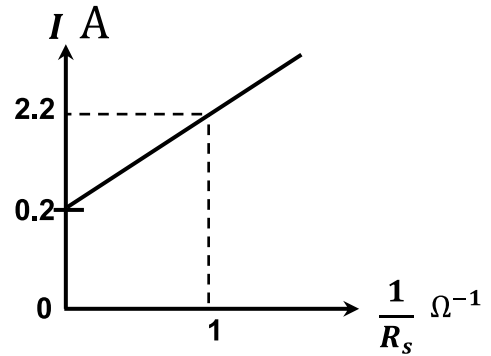
- في حالة التظليل على أكثر من رمز، تعتبر الإجابة خطأ.
 - في حالة ما إذا أجبنا إجابة خطأ، ثم قمنا بشطبها وأجبنا إجابة صحيحة تُحسب الإجابة صحيحة.
 - في حالة ما إذا أجبنا إجابة صحيحة، ثم قمنا بشطبها وأجبنا إجابة خطأ تُحسب الإجابة خطأ.
- ملحوظة:** يفضل عدم تكرار الإجابة على الأسئلة.
- عدد صفحات الكراسة (١٦) صفحة.
 - تأكد من عدد صفحات كراستك، فهي مسئوليتك.
 - زمن الامتحان (ثلاث ساعات).
 - الدرجة الكلية للامتحان (٦٠) درجة.
 - عدد الأسئلة (٥) لكل سؤال (١٢) درجة، لكل جزئية (درجة واحدة)
 - عند احتياج الطالب للإجابة على أي فقرة وذلك عند حدوث أي سبب يقتضي ذلك؛ يستخدم المسودة بآخر الورقة الامتحانية مع كتابة رقم السؤال والفقرة بوضوح، بشرط ألا تكون الإجابة مكررة.

هذا الجزء

غير مخصص للإجابة

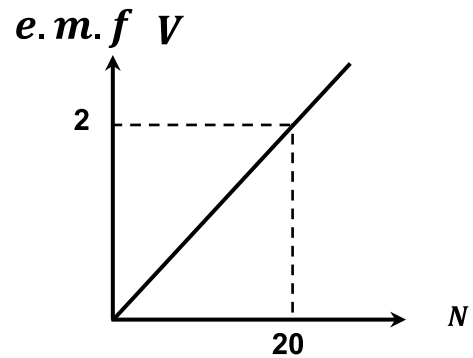
[First question]**(A) Choose the correct answer of each of the following:**

- 1- The opposite graph represents the relation between the maximum current **I** that can be measured and the reciprocal of the shunt resistance connected to a galvanometer. Then, the value of the maximum current that can be measured when the value of $R_s = 0.5 \Omega$ is



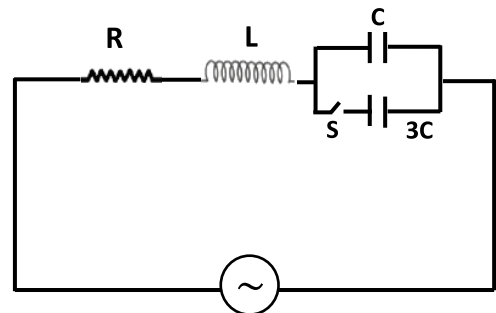
- (a) 3.4 A (b) 8 A (c) 14.2 A (d) 4.2 A

- 2- The opposite figure represents the graphical relation between the induced **emf** generated in a coil and the number of its turns. The coil is placed perpendicular to a magnetic field, and its cross-sectional area is 1m^2 . Therefore, the rate of change of the magnetic flux density is equal to.....



- (a) 0.1 T/s (b) 0.2 T/s (c) 0.3 T/s (d) 0.4 T/s

- 3- The figure represents an RLC circuit in the resonance condition. When switch **S** is closed, the value of the self-inductance of the coil that must be connected so that the circuit remains in resonance with the other coil is



- (a) $\frac{L}{3}$ parallel (b) series L (c) series $\frac{L}{4}$ (d) parallel L

- 4- In the electron microscope, if the minimum kinetic energy of the electrons used to observe an object of dimensions **2 nm** is **E** then the minimum kinetic energy of the electron that can be used to observe an object of dimensions **1 nm** is equal to.....

- (a) 2 E (b) $\sqrt{2}$ E (c) 4 E (d) 16 E

5- A copper wire carries a current of intensity **2A** when the potential difference across its ends is **20V** If it is stretched so that its length becomes double, then its resistance becomes:

- (a) 5Ω (b) 10Ω (c) 20Ω (d) 40Ω

6- The depletion region in a p–n junction contains:.....

- (a) Electrons only (b) Holes only
(c) Electrons and holes (d) Positive and negative ions

7- In the helium–neon laser, the metastable levels of neon and helium atoms are:

- (a) E2,E3 (b) E1,E3 (c) E1,E2 (d) E1,E2,E3

(B) First:

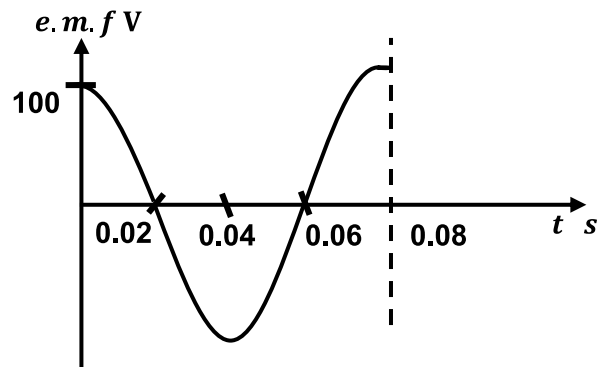
An AC dynamo whose coil resistance is negligible has its angular speed increased to double its original value.

- Choose the correct answer from the given answers for each of the following statements in each case.

1- The maximum EMF becomes:	(a)	Constant	(b)	doubles	(c)	becomes half	(d)	Zero
2- The periodic time:	(a)	Constant	(b)	doubles	(c)	becomes half	(d)	Zero
3- The average value of the EMF over one complete cycle becomes:	(a)	Constant	(b)	doubles	(c)	becomes half	(d)	Zero

Second:

The opposite figure represents the curve of the EMF generated in the coil of a resistance less dynamo during one complete cycle. It is connected in series with an inductor whose self-inductance is $\frac{1}{\pi}$. Calculate the current intensity passing through the coil.



.....

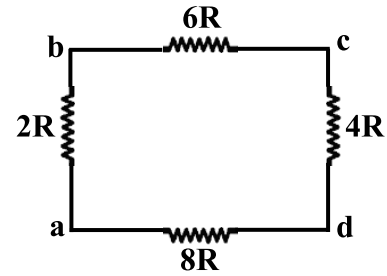
.....

.....

.....

[Second question]**(A) Choose the correct answer of each of the following:**

- 1- In the opposite figure, in order for equal current to pass through all resistors, the battery should be connected between points:



- (a) a,c (b) b,d
 (c) c,d (d) a,d

- 2- In the helium-neon laser, helium atoms are excited due to

- (a) the gas pressure inside the tube
 (b) the collision with neon atoms
 (c) the high continuous voltage
 (d) collision with the tube wall

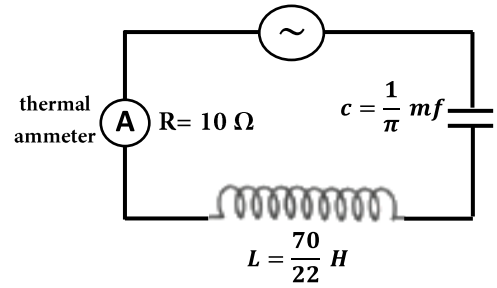
- 3- galvanometer whose coil resistance is 50Ω and whose maximum measurable potential difference is V . To increase its range by 10 times, its coil should be connected to resistance

- (a) 50Ω in parallel (b) 50Ω in series
 (c) 500Ω in series (d) 500Ω in parallel

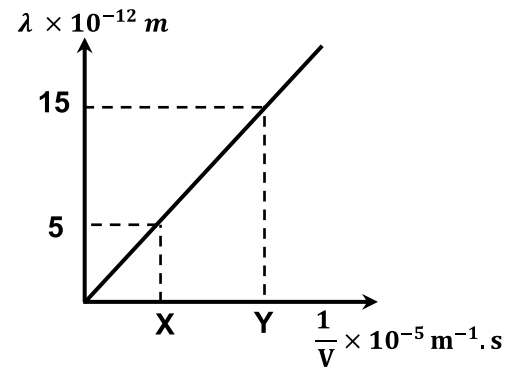
- 4- Two identical coils, one made of copper and the other of aluminum, $((\rho_e)_{Cu} < (\rho_e)_{Al})$. are exposed to the same rate of change of magnetic flux, then that in the aluminum coil.

- (a) the current intensity in the copper coil is greater than
 (b) the induced **emf** in the copper coil is less than
 (c) the induced **emf** in the copper coil is greater than
 (d) the induced current intensity in the copper coil equals

- 5- In the opposite figure, a coil of negligible ohmic resistance is connected with a capacitor, an AC source, and a thermal ammeter of resistance $10\ \Omega$. The reading of the thermal ammeter is $[\pi = \frac{22}{7}]$.



- 6- The opposite figure shows the relation between the wavelength associated with the motion of a particle and the reciprocal of its speed. Then the ratio between the momenta $\frac{(P_L)_x}{(P_L)_y} = \dots\dots\dots$ ($h = 6.625 \times 10^{-34}\ J.s$)



- (a) $\frac{1}{3}$ (b) $\frac{3}{1}$ (c) $\frac{1}{5}$ (d) $\frac{5}{1}$
- 7- The function of the prism in the spectrometer is..... .
- (a) produce parallel rays
 (b) analyze white light into its primary components
 (c) produce a pure spectrum
 (d) form a focus for each color (light) on its focal plane

- 1- Direction of the force acting on side **cb** is
- (a) upward (b) into to the page
- (c) out of the page (d) it is not affected by a force
- 2- Direction of the force affecting side **ab** is
- (a) upward (b) into to the page
- (c) out of the page (d) it is not affected by a force
- 3- Direction of the magnetic dipole moment is
- (a) upward (b) into to the page
- (c) out of the page (d) it is not affected by a dipole

$$(e = 1.6 \times 10^{-19}c, c = 3 \times 10^8 m/s, h = 6.625 \times 10^{-34} J.S)$$

1) The energy of the level to which the electron has transitioned.

[Third question]**(A) Choose the correct answer of each of the following:**

1- In the Compton effect, the quantity that increases after the collision for the photon is

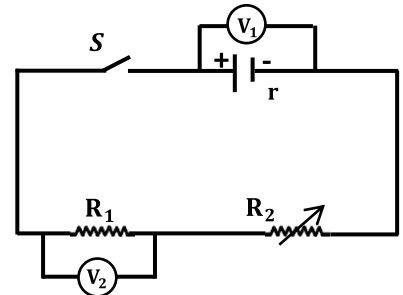
- (a) speed
- (b) momentum
- (c) energy
- (d) radius

2- If the concentration of phosphorus atoms in a silicon crystal is 10^{12} cm^{-3} and the concentration of holes in it is 10^8 cm^{-3} , then the concentration of electrons in the silicon crystal is

- (a) 10^8 cm^{-3}
- (b) 10^{10} cm^{-3}
- (c) 10^{12} cm^{-3}
- (d) 10^{20} cm^{-3}

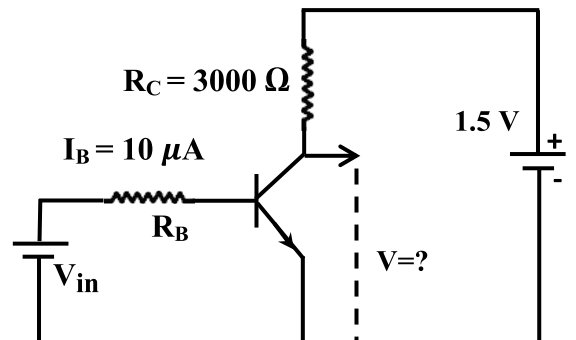
3- In the opposite figure, when the switch is closed, when R_2 is increased, the reading of the device V_1 is

- (a) decreases first, then increases
- (b) decreases and does not reach zero
- (c) increases
- (d) decreases until it reaches zero



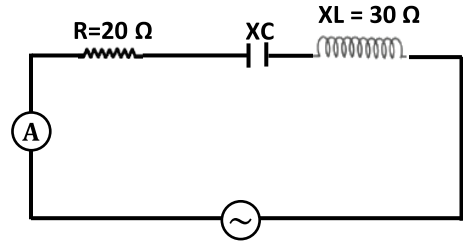
4- The opposite figure represents a transistor. Using the data shown in the figure, if the current gain $\beta_e = 40$, then the value of $V = \dots$

- (a) 0.1 V
- (b) 0.2 V
- (c) 0.3 V
- (d) 0.4 V



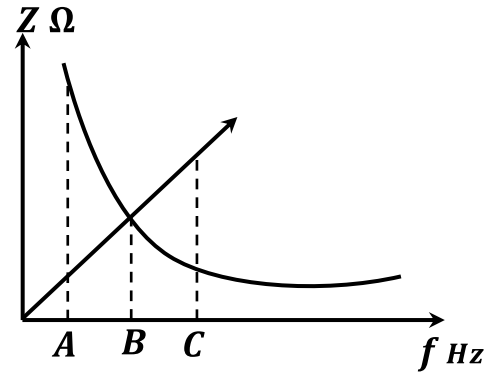
5- In the opposite figure (**LCR circuit**), if the phase angle is (-45°) , then the value of $X_C = \dots\dots\dots$

- (a) 10Ω (b) 20Ω
(c) 40Ω (d) 50Ω



6- The opposite graph represents the relation between the inductive reactance X_L , the capacitive reactance X_C , and the source frequency f . Then the ratio ... at point $\frac{X_L}{X_C}$ is $\dots\dots\dots$

- (a) greater than 1 (b) equal to 1
(c) less than 1 (d) equal to $\frac{1}{2}$

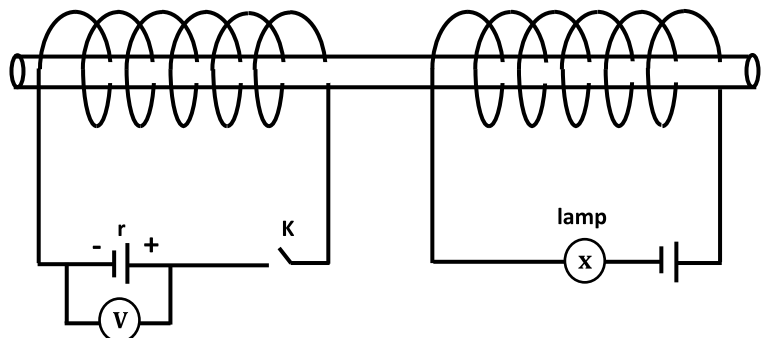


7- A solenoid carries a current of intensity I and its internal resistance is negligible. The magnetic flux density at its axis is B . If one-third of the coil is cut off and the same current passes through it, then the magnetic flux density at the axis becomes $\dots\dots\dots$

- (a) $\frac{B}{3}$ (b) B (c) $3B$ (d) $9B$

(B) First:

The following figure represents two coils wound around a soft iron core. What happens to each of the following when switch K is closed? Give a reason:



1- The illumination of lamp X .

.....

2- The voltmeter reading.

.....

Second:

The following table represents the relation between the **emf** generated in the dynamo coil and the angle of rotation θ

emf	0	$60\sqrt{2}$	120	$60\sqrt{2}$	0
angle of rotation θ	0°	45°	90°	135°	180°

1- Plot the graph of **emf** on the vertical axis against the angle of rotation θ on the horizontal axis.

2- From the graph, calculate the effective value.

.....

.....

.....

.....

3- Calculate the average induced EMF during the interval of rotation of the coil from 0° to 45° .

.....

.....

.....

.....

[Fourth question]**(A) Choose the correct answer of each of the following:**

1- A laser light source is placed at a certain distance from the cathode of a photoelectric cell, causing electrons to be emitted. If the distance between the source and the cathode is reduced to half, then:

.....

- (a) the kinetic energy of the emitted electrons increases
- (b) the number of emitted electrons decreases
- (c) the work function of the cathode surface increases
- (d) the number of emitted electrons remains constant

2- An external resistance of $200\ \Omega$ is connected across an ohmmeter, and its pointer deflects to $\frac{3}{4}$ of the scale whose full-scale value is $5\ \text{mA}$. Then the **emf** of the battery is:

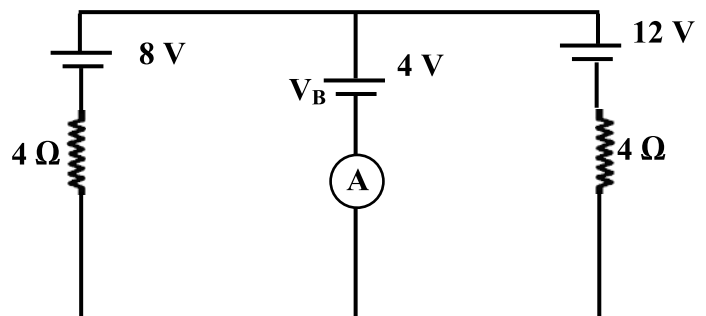
- (a) $1.5\ \text{V}$ (b) $2\ \text{V}$ (c) $2.5\ \text{V}$ (d) $3\ \text{V}$

3- If the split-ring (commutator) in a motor is replaced by slip rings, the coil:

- (a) continues rotating in the same direction
- (b) remains fixed and does not rotate
- (c) oscillates about the vertical position
- (d) rotates and then stops after a period

4- From the opposite figure, using Kirchhoff's laws, the ammeter reading is

- (a) $1\ \text{A}$ (b) $2\ \text{A}$
 (c) $3\ \text{A}$ (d) $4\ \text{A}$

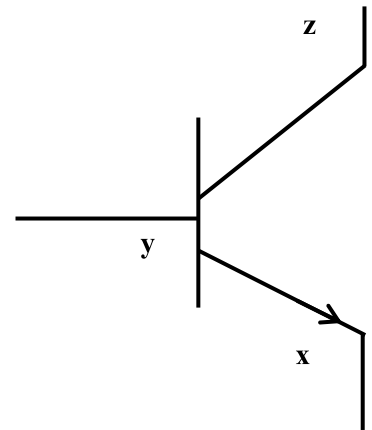


5- The function of the cylinder fixed inside the moving-coil galvanometer is :

- (a) increasing the range of the instrument
- (b) increasing the sensitivity of the instrument
- (c) reducing friction
- (d) making the flux constant during rotation

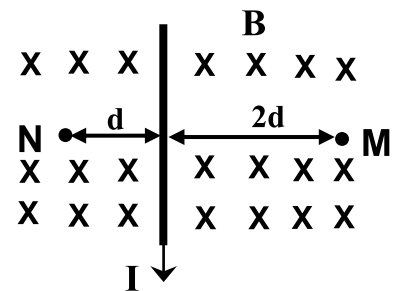
6- The opposite figure is the symbol of a transistor. For it to operate as an ON switch, it must be potential:

- (a) X positive, Y negative, Z negative
- (b) X negative, Y positive, Z positive
- (c) X positive, Y positive, Z negative
- (d) X negative, Y negative, Z negative



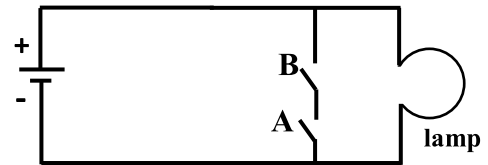
7- In the opposite figure, the straight wire carries an electric current I and is placed in a uniform magnetic field of flux density B . If the magnetic flux density at point N is $3B$, then the magnetic flux density at point M is

- (a) B
- (b) $2B$
- (c) $3B$
- (d) 0

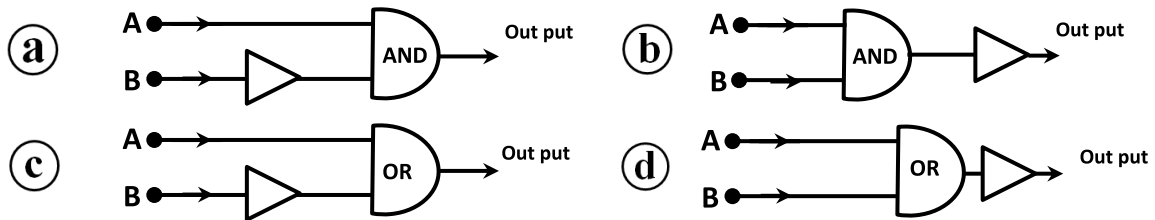


(B) First:

The opposite electric circuit is equivalent to logic gates:



1- Choose from the following figures the gates that represent it.

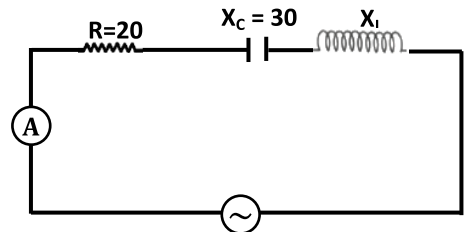


2- Complete the corresponding truth table.

A	B	Out put
0	0	
1	0	
1	1	

Second:

In the opposite figure, the circuit is in resonance. The ammeter reading is **2 A**, and the power lost in the circuit is **240 W**. From the data shown, calculate:



1- The potential difference across the coil.

.....

.....

.....

.....

2- The phase angle.

.....

.....

.....

.....

[Fifth question]**(A) Choose the correct answer of each of the following:**

1- In Faraday's experiment, the direction of the induced current depends on:

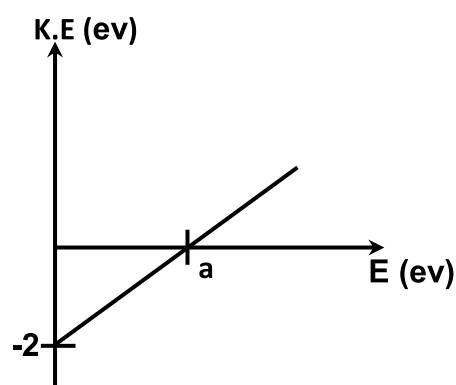
- (a) the speed of the magnet entering
- (b) the strength of the magnet
- (c) the type of the magnetic pole facing the coil
- (d) the number of turns of the coil

2- A circular coil of **N** turns is connected to a battery of negligible internal resistance. If the number of turns is doubled while the radius of the coil remains constant, and it is connected to the same battery, then the magnetic flux density at its center

- (a) remains constant
- (b) becomes half
- (c) doubles
- (d) becomes four times

3- The opposite graph represents the relation between the kinetic energy of an electron emitted from the cathode of a photoelectric cell and the energy of the incident photon. The value of the critical frequency is..... Hz .

- (a) 2.4×10^{14}
- (b) 2.4×10^{15}
- (c) 4.8×10^{14}
- (d) 4.8×10^{15}

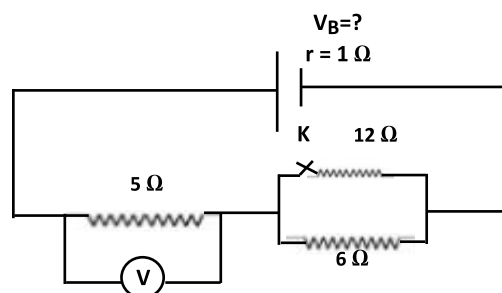


4- In the Coolidge tube, if the potential difference between the filament and the target increases, then:

- (a) the minimum wavelength of the continuous spectrum decreases
- (b) the minimum wavelength of the continuous spectrum increases
- (c) the wavelength of the characteristic spectrum decreases
- (d) the wavelength of the characteristic spectrum increases

5- In the opposite figure, if the voltmeter reading is **15 V**, then its reading when the switch is opened becomes:

- (a) 10 V
- (b) 12.5 V
- (c) 20 V
- (d) 30 V



6- In holography, the reference rays and the rays reflected from the illuminated object are both:

- (a) the same in wavelength, intensity, and phase
- (b) the same in wavelength but different in intensity and phase
- (c) different in wavelength, intensity, and phase
- (d) different in wavelength but the same in intensity and phase

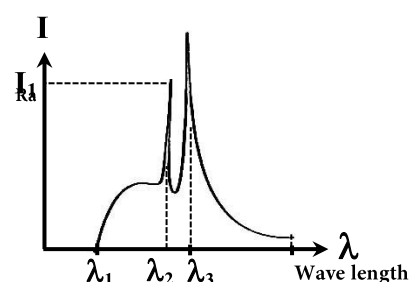
7- The eddy currents induced increasing in a metallic cylinder wrapped around it a coil carrying **AC** flows when:

- (a) the cross-sectional area of the coil increases
- (b) the resistivity of its material increases
- (c) the electrical conductivity of its material increases
- (d) the frequency of the current passing through the coil decreases

(B) First: The opposite figure represents the

X-ray spectrum emitted from a Coolidge tube.

What happens to each of λ_1 , λ_2 , λ_3 and I_1 in the following cases:



1- When the filament current intensity increases.

2- When the target material is replaced by an element of greater atomic number.

Second: A step-down transformer of efficiency **80%** loses energy at a rate of **200 J/s**. Given that the potential difference across the primary coil is **2000 V**, and the turns ratio of its two coils are $\frac{5}{2}$.

Calculate the current intensity in the secondary coil.

Draft